



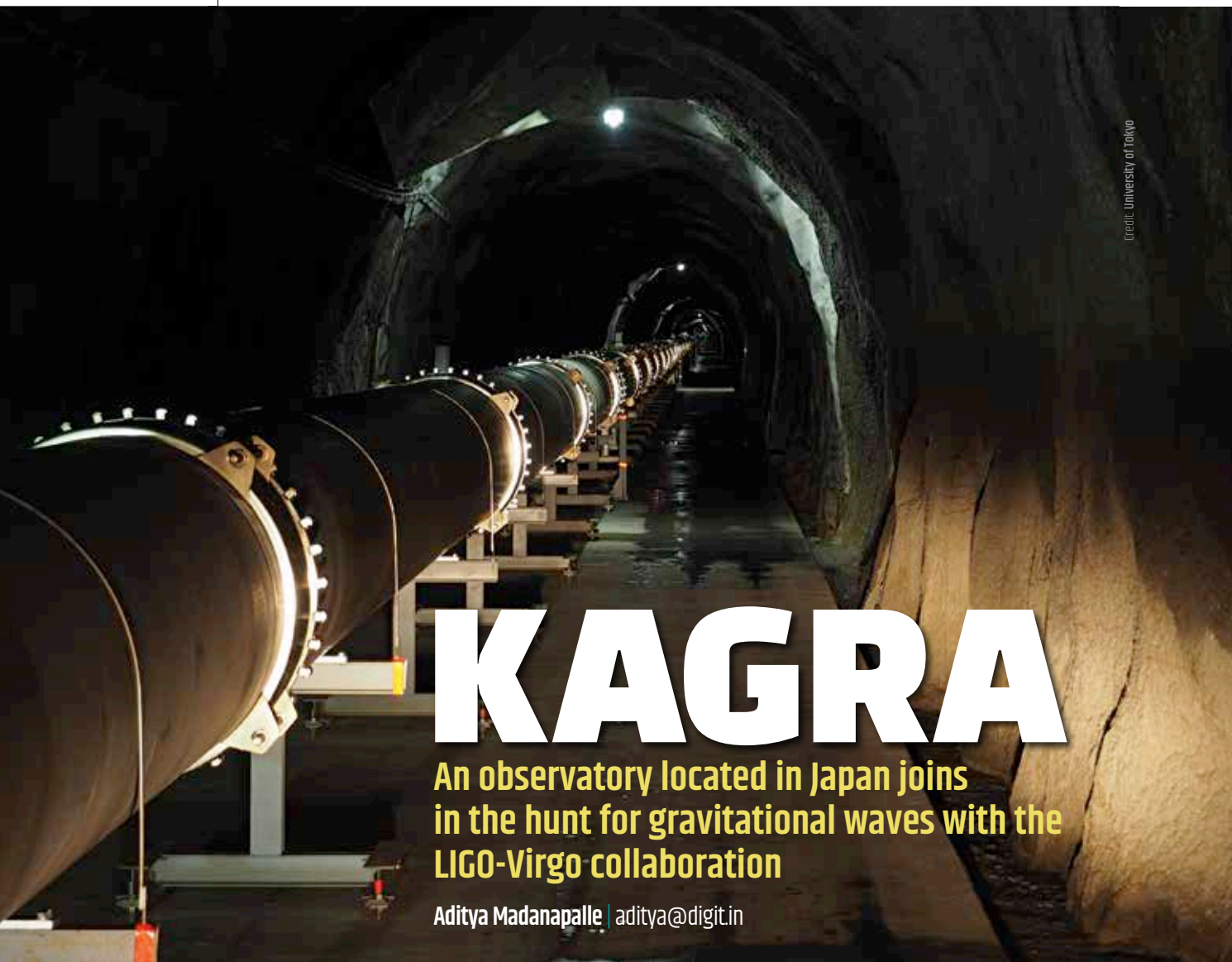
Closer to the Moon

NASA contractors are building on the agency's space shuttle for Artemis Moonshot scheduled in 2024. <https://digit.in/nov19-39>



Crashing exoplanets

Astronomers witnessed the aftermath of a crash between two exoplanets surprising late into their lives. <https://digit.in/nov19-40>



Credit: University of Tokyo

KAGRA

An observatory located in Japan joins in the hunt for gravitational waves with the LIGO-Virgo collaboration

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n October 4, the work on the Kamioka Gravitational Wave Detector (KAGRA) in Japan was complet-

ed, with the construction of an underground cryogenic chamber. Work on the project began in 2010, and took most of the decade. The project was supposed to be finished by 2017, but was delayed by two years. The detector is located in an abandoned mine in the Gifu prefecture of Japan, and is the first major gravitational wave observatory in Asia. KAGRA is the fourth major gravitational wave detector in the world, after the two Laser Interferometer Gravitational-Wave Observatory (LIGO) detectors in the US built and operated by Caltech and MIT, and the Virgo interferometer in Italy.

The distinguishing feature of the KAGRA detector is that it is the first major gravitational wave detector to use cryogenic mirrors. These mirrors are made of sapphire crystals that help reduce thermal noise. The underground site is at a seismically-quiet location, which is also advantageous to the detector. The KAGRA detector will be the first kilometre scale detector to be housed underground, as well as the first to use cryogenically-cooled mirrors. The successful demonstration of these novel technologies will help

provide direction to more powerful detectors in the future. Takaaki Kajita, the principal investigator of the KAGRA project said, "we should make every effort, for the global gravitational-wave community, to prove that the underground site and the cryogenic mirrors are useful." Jo van den Brand, Virgo Collaboration Spokesperson said, "with KAGRA joining, gravitational-wave science will become a global collaborative effort. The Virgo Collaboration looks forward to learning from the new and innovative approach of utilising an underground and cryogenic interferometer."

As soon as the construction was finished, a memorandum of under-